Python for Beginners: All About Variables!

Welcome back! In the last worksheet, you learned how to print words to the screen and how to ask the user for input. You also got a sneak peek at variables. Today, we are going to dive a little deeper into what variables actually are and the different types of information we can store inside them!

Setting up your environment

1. Just like last time, go to Google Colab: https://colab.research.google.com/

2. Click "File", then "New Notebook in Drive".

3. Get ready to type in the empty code blocks! If you have any problems getting set up, put up your hand and ask an educator for help.

What exactly is a variable?

Think of a variable as a labeled box where you can store information to use later.

If you are playing a video game, the game needs to remember your character's name, your score, and how many lives you have left. The game programmers use variables to remember that information!

Let's create a variable for a game character. Copy and paste this code into Colab and click the play button:

```
character_name = "Mario"
print(character_name)
```

Output:

```
Mario
```

Here is what happened: we created a "box" (variable) called character_name, and we put the word "Mario" inside it. When we told the computer to print character_name, it looked inside the box and printed out what it found.

Strings (Text)

In programming, we don't just call text "words" or "text." We call it a **String**.

A String is just a bunch of characters strung together. The most important rule to remember about Strings is that **they must always be inside quotation marks (" ")**.

Try running this code:

```
secret_message = "Python is awesome!"
print(secret_message)
```

Because the words are inside quotation marks, the computer knows this is a String variable.

Integers (Numbers)

What if we want to store a number, like a high score? In Python, whole numbers are called **Integers**.

The most important rule about Integers is that **they do NOT use quotation marks**.

Try running this code:

```
player_score = 100
print(player_score)
```

Because there are no quotation marks around 100, the computer knows this is an Integer (a number), and not a String (text).

Why does the difference matter?

You might be wondering: *"Why do I care if something is a String or an Integer? Don't they both just print out the same way?"*

That's a great question! It matters because the computer treats text and numbers very differently, especially when we want to do some math.

Let's look at a trick. Run these two lines of code in Colab and look closely at the output:

```
print(5 + 5)
print("5" + "5")
```

Output:

```
10
55
```

Wait, what happened there?!

- **print(5 + 5)**: Because there are no quotation marks, the computer knows these are Integers. It does the math and adds them together to get 10.
- **print("5" + "5")**: Because there ARE quotation marks, the computer thinks these are Strings (just pieces of text). Instead of doing math, it just glues the two pieces of text together side-by-side to make "55".

The best way to learn programming is to mess around and tinker with things! Try changing the numbers and the text to see what happens.

Exercises:

Time to test out your new skills! Try writing the code for these challenges:

1. Create a variable called my_favourite_color and store a **String** inside it. Print it out to the screen.
2. Create a variable called number_of_pets and store an **Integer** inside it (the number of pets you have, or how many you want!). Print it out to the screen.
3. Create two integer variables (for example: score_1 = 10 and score_2 = 20). Then, try to print(score_1 + score_2).

Before you move on, put your hand up and ask an educator if you think you solved these exercises